

Predicting Term Deposit Subscription

A Decision Tree and Random Forest Comparison on Bank Telemarketing Data

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Overview

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The Dataset

UCI Bank Marketing data — 4,119 clients, 21 variables, imbalanced target

02

Preparation

Cleaning, feature engineering, and encoding

03

Exploration

What drives subscription — including contact timing

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Modelling

Decision Tree vs. Random Forest

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Findings

Results, key drivers, and what they mean for the bank

The Dataset

UCI “Bank Marketing” dataset (Moro, Cortez and Rita, 2014) — not sourced from Kaggle

4,119

Client records

21

Original variables

10.9%

Subscribed (“yes”)

0

Duplicate rows

Four variable groups:

- Client attributes — age, job, marital status, education, default/housing/loan status
- Current campaign — contact type, month, day of week, number of contacts
- Previous campaign — days since last contact, prior outcome
- Macroeconomic context — Euribor rate, employment level, price & confidence indices

Target is heavily imbalanced: 89.1% did not subscribe vs. only 10.9% who did — this shaped every modelling choice that followed.

Preparing the Data

Cleaning, feature engineering, and encoding

Cleaning

- No duplicate rows found
- “unknown” kept as a genuine category (not imputed) — up to 19.5% of values in ‘default’
- pdays sentinel (999 = never contacted) affects 3,959 of 4,119 rows

Feature Engineering

- was_contacted_before — binary flag extracted from the pdays sentinel
- age_group — six ordinal bands (18-25 ... 66+) from raw age

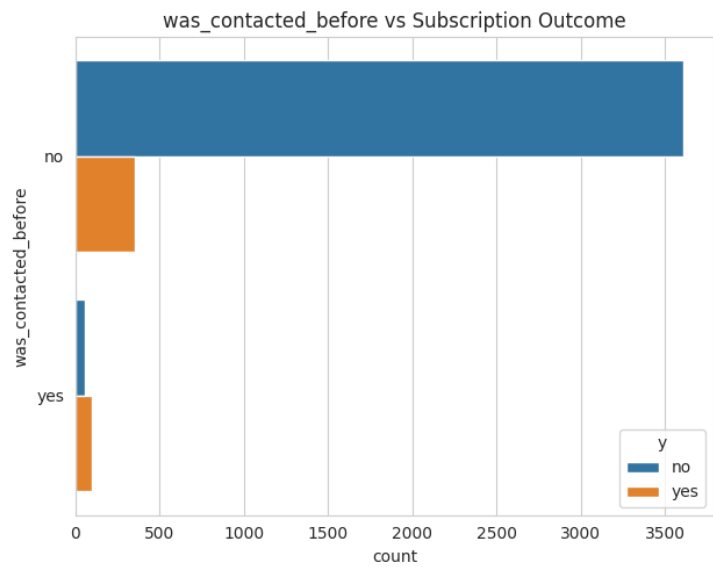
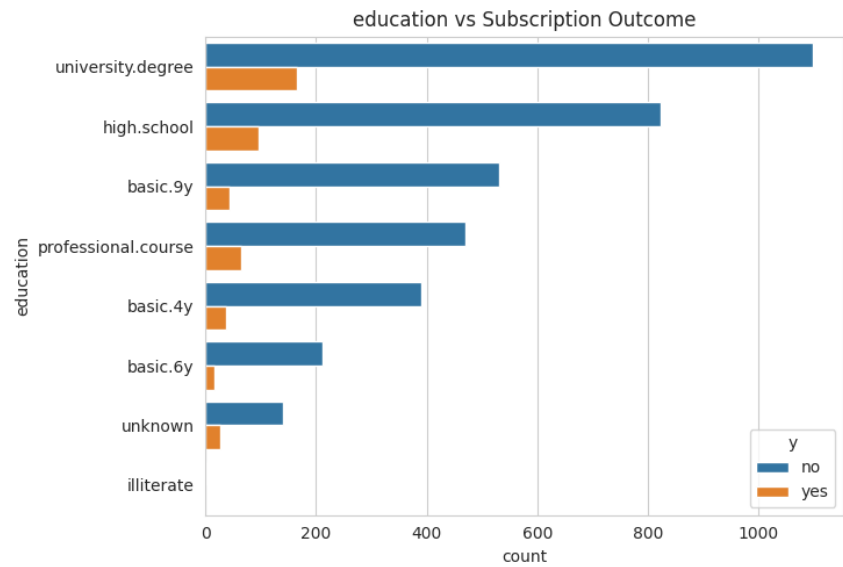
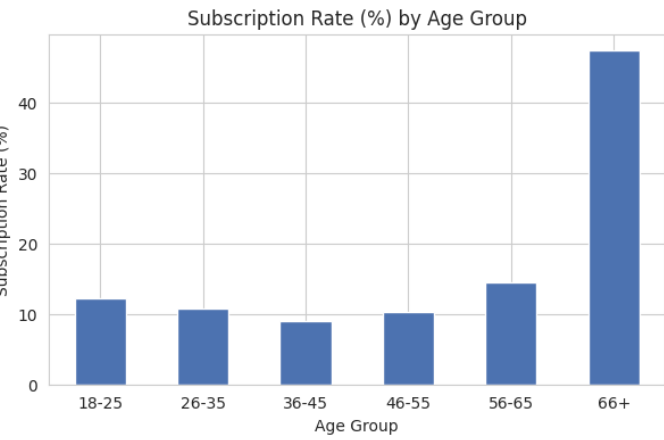
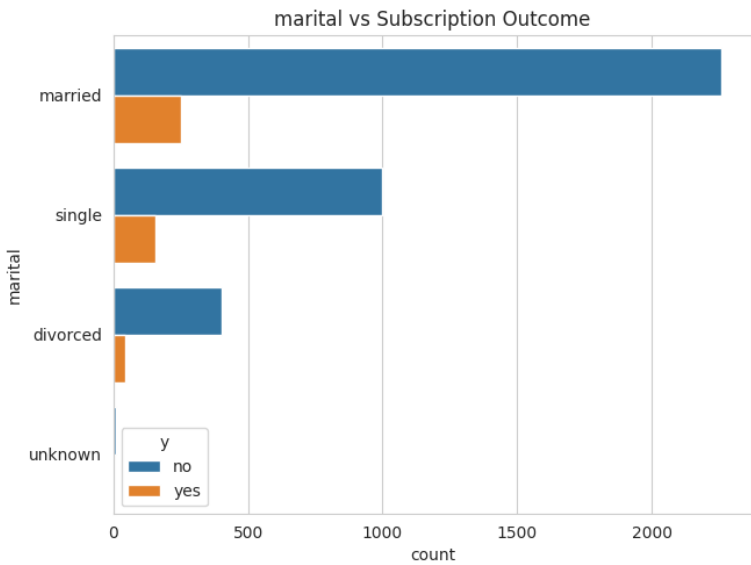
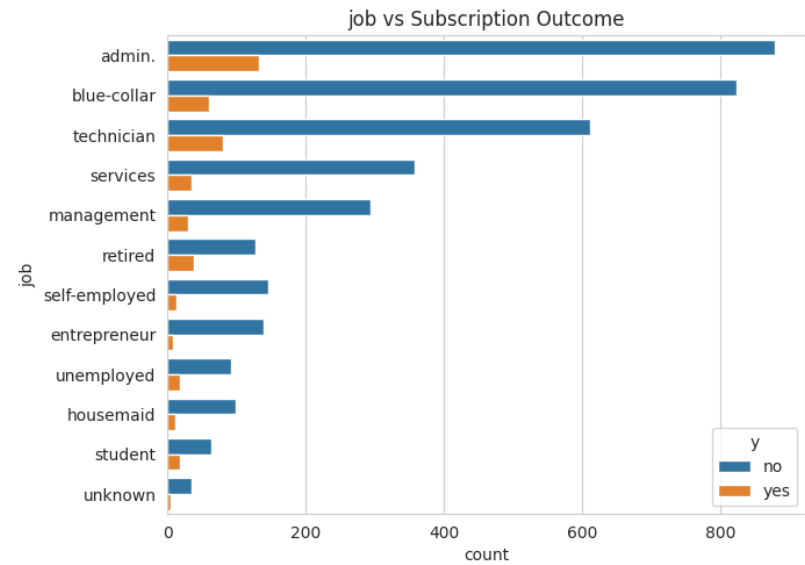
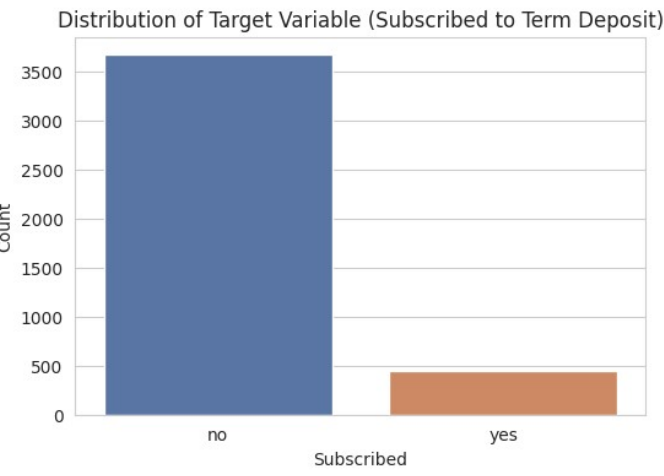
Encoding & Leakage

- 11 nominal variables → one-hot encoded (job, month, poutcome, etc.)
- age_group → ordinal integer encoding (preserves order)
- **duration dropped — only known after the call ends, so it leaks the outcome**

23 columns → 55 columns after encoding

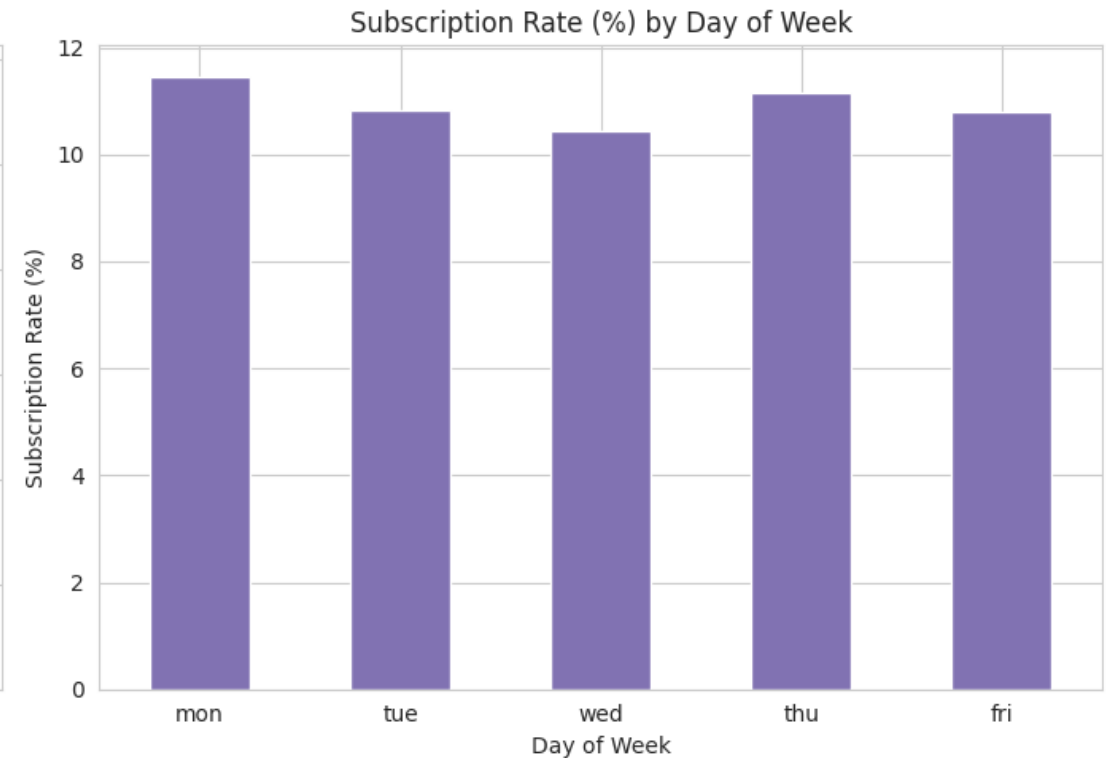
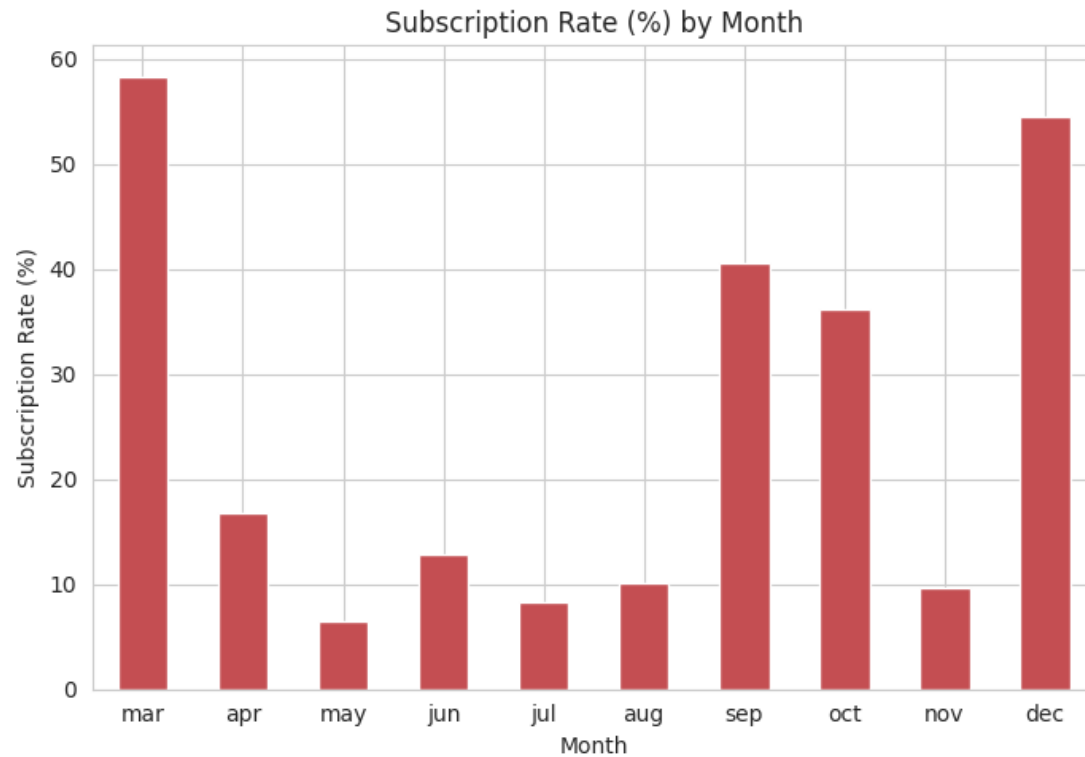
Who Subscribes?

Target distribution and key categorical patterns



When Does It Work Best?

Subscription rate by month vs. day of week



March, September, October and December convert far better (36-58%) than the busy summer months (6-13%) — despite receiving fewer calls. Day of week barely moves the needle.

Modelling Approach

Two tree-based classifiers, compared head-to-head

Decision Tree

- Single interpretable tree
- Max depth = 6
- class_weight = “balanced”
- Baseline model

Random Forest

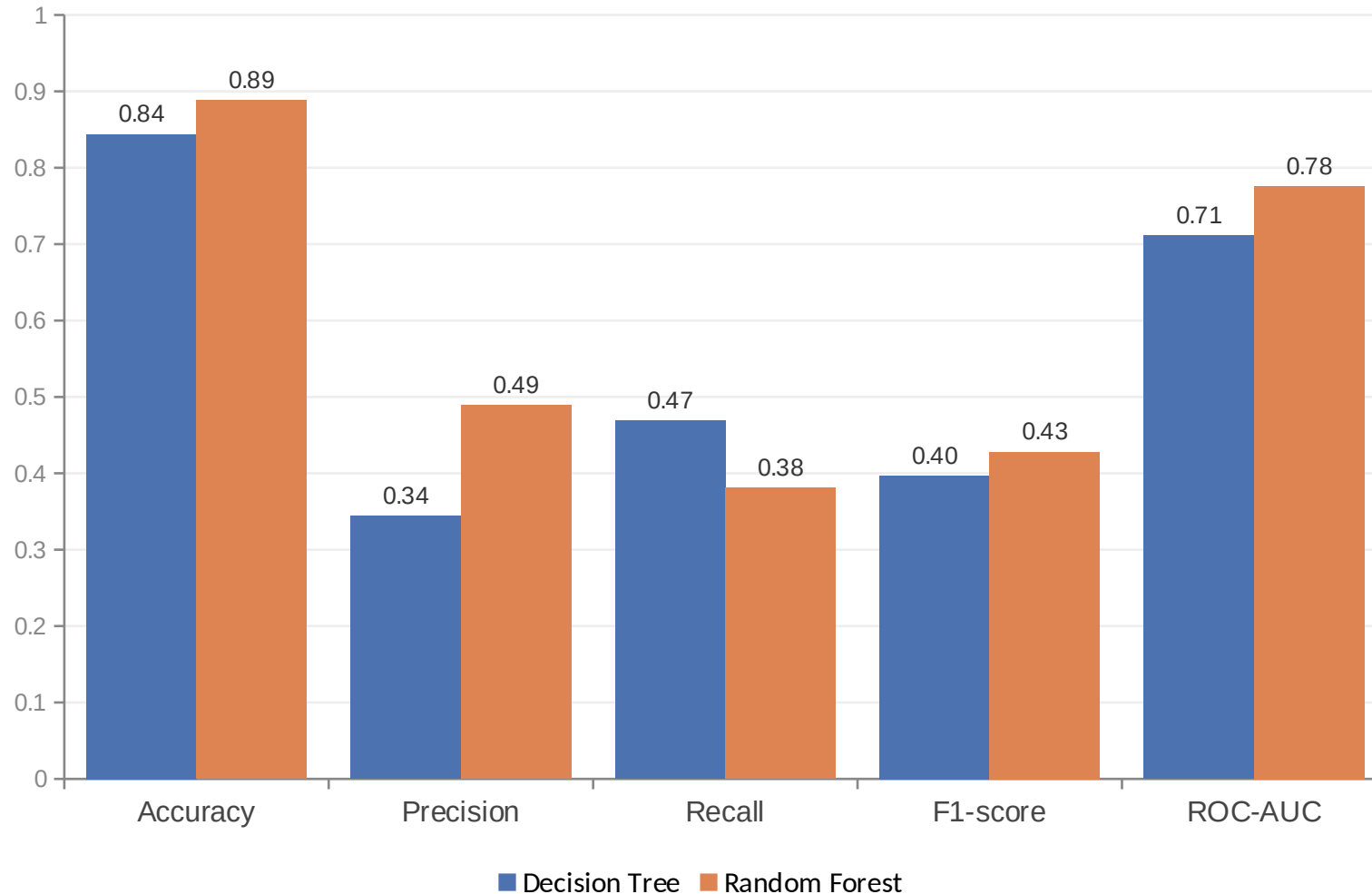
- Ensemble of 300 trees
- Max depth = 10
- class_weight = “balanced”
- Averages many decorrelated trees

Both trained on a 75/25 stratified train/test split (3,089 / 1,030 rows), preserving the 89/11 class balance in both sets.

Evaluated on accuracy, precision, recall, F1-score and ROC-AUC — not accuracy alone, given the class imbalance.

Model Performance

Random Forest outperforms the single Decision Tree on 4 of 5 metrics



Random Forest

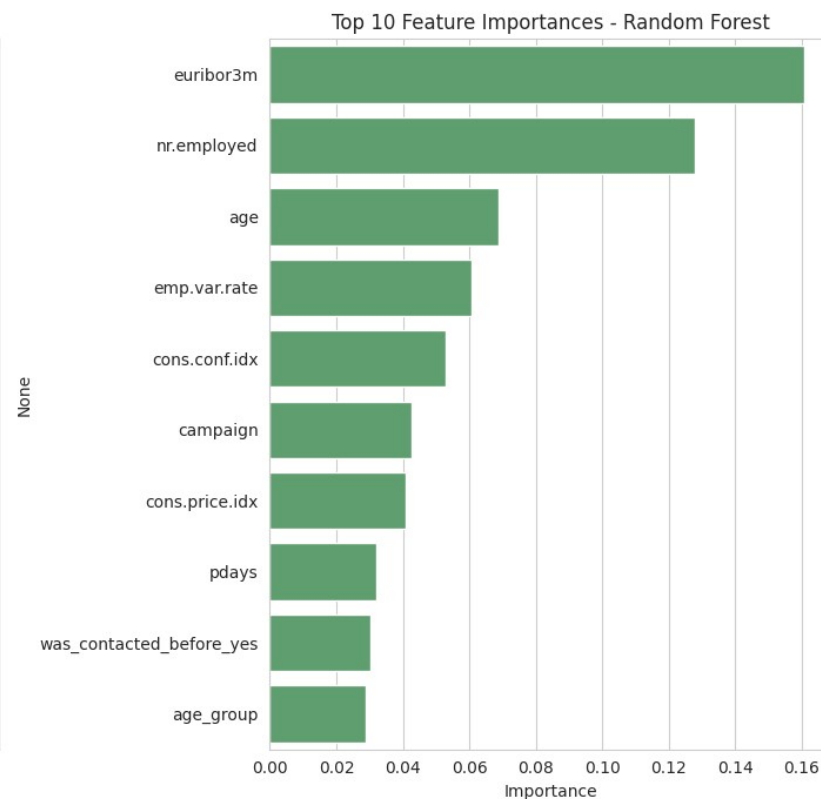
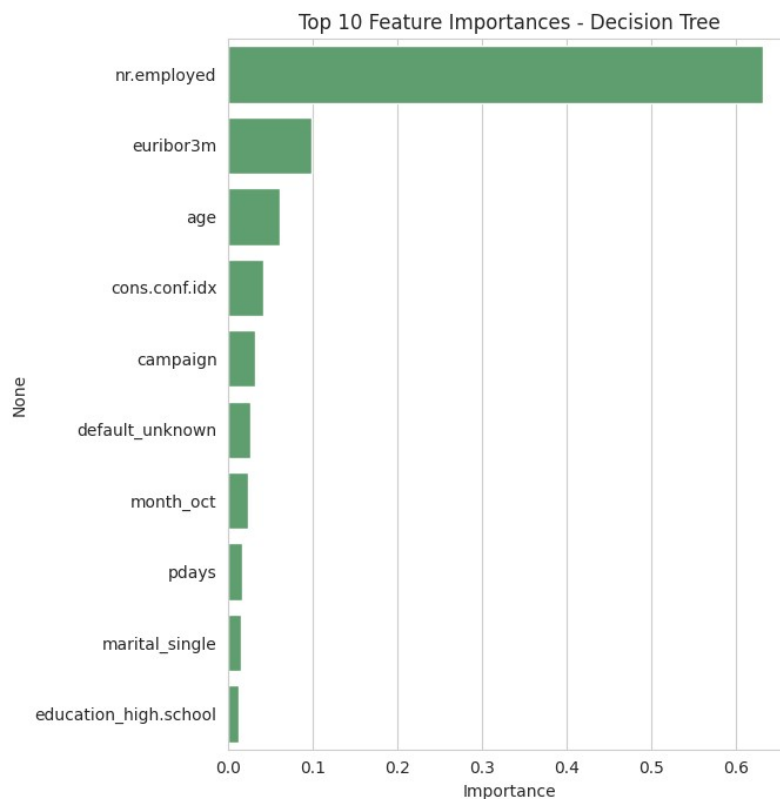
- 88.8% accuracy
- 0.776 ROC-AUC
- Fewer false positives

Decision Tree

- 84.4% accuracy
- Higher recall (0.469)
- More false positives

What Drives Subscription?

Top predictors from the Random Forest



euribor3m and nr.employed — both macroeconomic indicators — are the two strongest predictors, ahead of any individual customer attribute (~29% of total importance combined).

Key Insights

1

Timing beats targeting

Economic conditions at time of contact matter more than who is called — euribor3m and nr.employed dominate feature importance.

2

Ensembles generalise better

Random Forest's averaging across 300 trees reduces variance vs. a single Decision Tree, lifting accuracy from 84.4% to 88.8%.

3

Imbalance is the core challenge

Even with class-weighting, recall on subscribers stays at 0.38-0.47 — a real trade-off for deployment.

4

Leakage matters

Dropping duration keeps the model realistic for pre-call prediction, at some cost to raw accuracy.

Conclusion

- Random Forest is the stronger model overall (88.8% accuracy, 0.776 ROC-AUC) but the precision/recall trade-off vs. the Decision Tree should guide the final choice based on business priorities.
- Campaign timing and macroeconomic context are at least as valuable a lever for the bank as customer targeting.
- Next steps: threshold tuning or resampling (e.g. SMOTE) to improve recall on the minority “yes” class.

Thank you — Questions?

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